

1 Chapter 3.1

Problem 2. How many integers in $[100]$ are not divisible by 4, 6, or 7?

Answer:

$$\begin{aligned} N_{\geq}(\emptyset) &= 100 & N_{\geq}(d_4, d_6) &= \left\lfloor \frac{100}{4 \cdot 6} \right\rfloor = 4 \\ N_{\geq}(d_4) &= \left\lfloor \frac{100}{4} \right\rfloor = 25 & N_{\geq}(d_4, d_7) &= \left\lfloor \frac{100}{4 \cdot 7} \right\rfloor = 3 \\ N_{\geq}(d_6) &= \left\lfloor \frac{100}{6} \right\rfloor = 16 & N_{\geq}(d_6, d_7) &= \left\lfloor \frac{100}{6 \cdot 7} \right\rfloor = 2 \\ N_{\geq}(d_7) &= \left\lfloor \frac{100}{7} \right\rfloor = 14 & N_{\geq}(d_4, d_6, d_7) &= \left\lfloor \frac{100}{4 \cdot 6 \cdot 7} \right\rfloor = 0 \end{aligned}$$

The answer is $100 - (25 + 16 + 14) + (4 + 3 + 2) - 0 = 36$

Problem 6. Answer the hat-check problem (i.e., the problem of derangement) for general n . This number is known as D_n .

Answer:

Problem 7. The number D_n is the hat check problem for general n .

- (a) Calculate $\lim_{n \rightarrow \infty} \frac{D_n}{n!}$
- (b) Prove that for any n , D_n equals the closest integer to $n!/e$.

Answer:

Problem 8. How many ways can you distribute 20 identical objects to 10 distinct recipients so that each recipient receives at most five objects?

Answer:

Problem 9. How many ways can you distribute k identical objects to n distinct recipients so that each recipient receives at most r objects?

Answer:

Problem 17. A taxi drives from the intersection labeled A to the intersection labeled B in the grid of streets shown on page 94. The driver can only drive north (up) or east (right). Traffic reports indicate that there is heavy congestion at the intersections identified. How many routes from A to B can the driver take that ...

- (a) Avoid all the congested intersections?
- (b) Pass through at most one congested intersection?

Answer: Start by defining the different types of points. Ones that can go up and right. Ones that can only go up, ones that can only go right. Use the knowledge that each point has two choices (up or right) with a few exceptions.